HANUMANTHA RAO VULLI

Academic CV | AI Security, Governance and Reliable Data
Denton, Texas | vullihanumantharao@gmail.com
Portfolio: https://sunnyvulli.github.io/ | GitHub: https://github.com/Sunnyvulli
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EDUCATION

Master of Science in Data Engineering | University of North Texas | December 2024
Bachelor of Technology in Computer Science and Engineering | 2019 - 2022
KKR and KSR Institute of Technology and Sciences, Guntur
Diploma in Mechanical Engineering | 2016 - 2019
Kallam Haranadhareddy Institute of Technology, Guntur
Secondary School Certificate (SSC) | 2015 - 2016
Oxford Concept High School, Guntur

RESEARCH INTERESTS

Security and governance of agentic AI; evidence-supported decisions, data lineage, and reproducible evaluation. Current work examines citation support, bounded tool permissions, and approval-gated data workflows. Learned-agent security evaluation remains a future direction.

SELECTED RESEARCH AND ENGINEERING

Applied ML Research Suite | October 2026 | Executed research baselines
https://github.com/Sunnyvulli/Data-Engineer_AI

- Citation support: 2,604 ALCE sentences and 900 independent RAGTruth answers; source-only calibration, clustered uncertainty and failure analysis.
- Phishing robustness: grouped UCI/raw-URL experiments and 7,050 screenshot/URL pairs; compact image fusion reduced ranking performance.
- Early transient ranking: RAPID-inspired GRU and forest comparison on simulated PLAsTiCC light curves, with a shared three-day cutoff.
- Includes 58 integrity tests, checksums, batch scoring, measured outputs, and documented limits. No publication acceptance or production validation claimed.

Lakehouse Guardian and Catalog Sentinel | October 2026 | Local reference systems
https://github.com/Sunnyvulli/lakehouse-guardian

https://github.com/Sunnyvulli/catalog-sentinel

- Spark/Iceberg, dbt and Airflow data workflow with LangGraph approval gates: 21 controlled batches, row precision 1.00 and recall 0.80; plausible count inflation remains undetected.
- Dimensional mart: 17,379 hourly facts and 731 observed dates; 25 dbt data checks and 13 Python tests. Full-refresh local models, not CDC or a cloud warehouse.
- Read-only MCP tools: 14 behavioral tests; 13 expected denials enforced in 16 requests using actual stdio sessions. Host-assigned permissions, not remote authentication.
- Local execution and green CI; Terraform S3/Glue configuration validated, not deployed to AWS. No learned planner or model-level attack benchmark.

Evidence Ledger RAG | October 2026 | Extractive retrieval reference
https://github.com/Sunnyvulli/evidence-ledger-rag

- Incremental indexing, hybrid lexical retrieval and source-hashed extracts; 18 tests. On 14 authored queries, expected-file hit@3 was 11/14 versus BM25 10/14; no independent relevance labels.
- Lexical support triage flags 3/5 controlled unsupported claims and misses two meaning changes. Exact cited spans do not establish factual correctness.

Mobility Quality Lab | October 2026 | Technical portfolio
https://github.com/Sunnyvulli/mobility-quality

- Python/SQLite pipeline and read-only local API for 17,379 historical UCI Bike Sharing observations; data contracts, transactional loading, input hashes, and 13 passing tests.

UNPUBLISHED WRITING AND TEACHING

Length Baselines and Evidence-Support Triage: An ALCE-to-RAGTruth Audit | Manuscript draft for author review, October 2026; not peer-reviewed or accepted.

Data contracts and evaluation leakage in historical bike-rental analytics | Technical manuscript draft, October 2026; not peer-reviewed.

Reliable analytics from real observations | SQL/Python teaching sample with exercises, answer key, and assessment rubric.

PROFESSIONAL EXPERIENCE

Data Engineer | PRC Cloud Tech, USA | January 2026 - Present

AWS ingestion, ETL, reporting SQL, and troubleshooting services integrating Lambda, API Gateway, and RDS.

Data Engineer | Tech Force, USA | September 2025 - January 2026

AWS pipelines, PostgreSQL/Oracle query tuning, data-quality investigation, and CI/CD support.

Data Engineer | Infosys Now, USA | April 2025 - September 2025

Spring Boot services with AWS EC2, S3, and RDS; validation and workflow documentation.

Java Full Stack Developer | DBS Technologies, India | July 2022 - May 2023

Angular, Spring Boot REST APIs, AWS deployments, and Jenkins/Docker automation.

TECHNICAL SKILLS

Python, Java, SQL; Spark, Iceberg, dbt, Airflow, LangGraph, MCP; Spring Boot; PostgreSQL, Oracle; AWS EC2, S3, RDS, Lambda, API Gateway; Terraform, Docker, Jenkins, GitHub Actions.

CERTIFICATIONS

AWS Certified Data Engineer - Associate | Oracle Certified Professional: Java SE 17 Developer